

AWS Data Migration Tools Cheat Sheet

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Introduction to AWS Data Migration Tools

- **AWS Database Migration Service (DMS):** Facilitates seamless database replication with minimal downtime using change data capture and heterogeneous engine support.
- **AWS Snowball Family Devices:** Enables offline migration through rugged devices with capacities from 8TB to 100PB, ideal for large data transfers.
- **AWS Glue and Streaming Services:** Provides serverless ETL and streaming integration, automating schema discovery, transformation, and cataloging for analytics workflows.



AWS Database Migration Service (DMS)

- **Change Data Capture (CDC):** DMS leverages CDC to replicate ongoing database changes in real time, ensuring minimal latency during migration.
- **Advantages of DMS:** Supports multiple database engines with pay-as-you-go pricing, simplifying diverse and cost-efficient migration workflows.
- **Limitations and Constraints:** Limited ETL capabilities and reliance on network stability can impede complex transformations and continuous migration performance.

AWS Snowball & Snowcone Devices

- **Device Capacities and Use Cases:** Snowcone offers 8TB, Snowball Edge 80TB, Snowmobile 100PB for migrations and remote IoT data collection.
- **Security and Ruggedness:** Devices provide end-to-end encryption and rugged hardware suitable for harsh environments and secure transport.
- **Operational Considerations:** Shipping delays and logistics complicate timelines despite encryption advantages in offline bulk data transfers.

AWS Glue: Serverless ETL Service

- **Automatic Scaling and Integration:** Glue automatically scales resources, integrates seamlessly with AWS analytics, ensuring efficient ETL workflows at scale.
- **Advanced ETL Features:** Offers automatic code generation, schema discovery, and supports streaming ETL for dynamic data transformation needs.
- **Pros and Challenges:** Serverless convenience and multi-source support contrast with debugging difficulties and cold start latency issues.



Amazon S3 & Transfer Acceleration

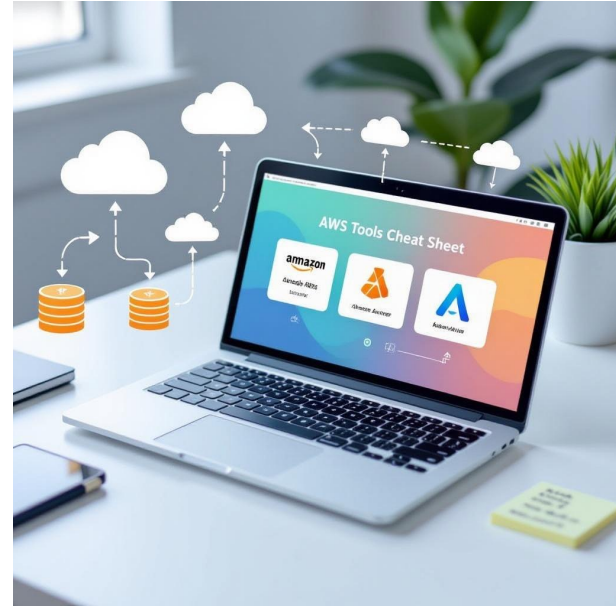
- **Scalable & Durable Storage:** S3 provides highly scalable object storage with 99.999999999% durability across multiple availability zones.
- **Transfer Acceleration Mechanism:** Transfer Acceleration uses CloudFront edge locations to optimize data uploads over long distances reducing latency.
- **Common Use Cases and Limitations:** Typical uses include data lakes, backup, static hosting; lacks native file system interface and incurs premium pricing.

Amazon Kinesis for Streaming Data

- **Kinesis Core Services:** Data Streams enables real-time ingestion; Firehose ensures reliable delivery; Analytics processes streams with SQL.
- **Key Use Cases:** IoT telemetry, application log streaming, and real-time fraud detection leverage Kinesis's low latency and scalability.
- **Operational Complexities:** Effective shard management and partition key design are critical for performance and scaling across distributed workloads.

Amazon RDS & Aurora: Managed Databases

- **Performance and Scalability:** Aurora offers up to 5x better performance than standard RDS with distributed, fault-tolerant storage.
- **Storage and Replication:** RDS supports single AZ and multi-AZ replication; Aurora uses shared storage with up to 15 replicas for high availability.
- **Pros and Limitations:** RDS is versatile with multiple engines; Aurora is cloud-optimized but less compatible outside AWS ecosystem and costlier.



Amazon Redshift for Analytics Workloads

- **Columnar Storage and MPP Performance:** Redshift's columnar storage and Massively Parallel Processing architecture enable fast large-scale OLAP query execution.
- **Seamless BI Tool Integration:** Supports native integrations with popular BI platforms like QuickSight, Tableau, and Power BI enhancing analytical workflows.
- **Architectural Limitations:** Lacks multi-AZ deployment; cold storage queries incur latency; optimized primarily for batch, not real-time, processing.

AWS Service Selection Guide

- **Data Migration Size & Tool Mapping:** Under 10TB with good network favors Direct Connect or DMS; 10-80TB with limited network uses Snowball or Snowcone.
- **Processing Needs & Corresponding Tools:** Kinesis supports real-time streaming; Glue manages batch ETL; DMS enables database migration; Redshift handles analytics and S3 storage.
- **Large Scale Migration Options:** Above 80TB migrations typically leverage Snowball Edge or Snowmobile for secure, offline high-volume data transport.

Real-Time E-commerce Data Pipeline Overview

- **Core Pipeline Challenges:** Balancing low latency, high update volumes, backpressure, and fault tolerance ensures reliable streaming analytics.
- **Key AWS Components:** DMS CDC captures changes; Kinesis Streams and Firehose manage data transport; Glue ETL transforms; Redshift analyzes; CloudWatch monitors.
- **Scenario Context:** Streaming PostgreSQL orders to S3, Glue, and Redshift enables near real-time analytics, improving e-commerce operational insights.



Real-Time Data Ingestion with AWS DMS

- **CDC from PostgreSQL via Logical Replication:** Utilizes PostgreSQL's logical decoding plugin wal2json to extract change events from transaction logs continuously.
- **DMS Replication Instance Functionality:** Acts as an intermediary capturing and transforming CDC stream from PostgreSQL logs to target endpoints efficiently.
- **Near-Zero Latency Streaming Enabled by DMS:** Streaming changes with minimal delay supports timely downstream analytics through optimized replication mechanisms.

Streaming Data Transport via Amazon Kinesis Data Streams

- **Scalable, Durable Real-Time Streaming:** Kinesis Data Streams handles high-throughput, real-time data streaming with durability and fault-tolerant storage.
- **DMS to Kinesis Data Flow:** DMS captures changes as CDC events and streams them directly into Kinesis shards for ordered delivery at scale.
- **Partitioning and Ordering Guarantees:** Shard-based partitioning enforces strict event ordering per shard, enabling parallel processing with backpressure control.

Data Delivery and Buffering Using Kinesis Data Firehose

- **Buffering and Delivery Mechanism:** Firehose buffers streaming data from Kinesis Streams with adjustable size and interval thresholds for efficiency.
- **Built-in Fault Tolerance:** Automatically retries delivery on failures, stores data locally if destinations are unreachable, ensuring data durability.
- **Auto-scaling and Transformation:** Dynamically scales throughput based on incoming data rate; supports optional real-time data transformations via AWS Lambda.



Real-Time ETL with AWS Glue Streaming Jobs

- **Continuous Data Transformation:** Glue Streaming jobs continuously ingest data from S3 or Kinesis, applying real-time schema mapping and validation.
- **Optimized Data Formats:** Transforms data into efficient columnar formats like Parquet, optimizing storage and query performance downstream.
- **Scalable Serverless Processing:** Glue automatically scales ETL jobs dynamically, handling fluctuating streaming volumes without manual intervention.

Analytics and Visualization with Amazon Redshift and QuickSight

- **Continuous Data Loading from S3:** Utilize COPY commands or Spectrum to load transformed data continuously from S3 into Redshift enabling near real-time analytics.
- **Redshift for Analytical Workloads:** Redshift's columnar storage and MPP architecture optimize performance for complex analytical queries on large datasets.
- **QuickSight for Business Intelligence:** QuickSight connects directly to Redshift, enabling creation of interactive dashboards supporting interactive drill-down and visual analytics.

Security, Monitoring, and Key Tradeoffs

- **Robust Security Controls:** Encryption via AWS KMS safeguards data at rest and in transit; IAM enforces fine-grained permissions; VPC and Security Groups enable strict network isolation.
- **Comprehensive Monitoring:** CloudWatch metrics and alarms monitor pipeline health; CloudTrail logs API activity; detailed component logs facilitate proactive troubleshooting and compliance.
- **Architectural Tradeoffs:** Increased system complexity and cost balance enhanced security and observability; schema evolution complicates migrations; requires data volume assumptions and skilled expertise.



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